EXPERIMENT – 10

AIM:

To develop an Android application that allows the user to change the font and color of text displayed on the screen and display a Toast message when a button is pressed.

ALGORITHM:

1. Initialize the Android Application:

o Set up a new Android project in Android Studio. o Create a MainActivity Java file for the main screen of the application.

2. Design the User Interface (UI):

o In the XML layout file, add the following UI components:

TextView: For displaying the text whose font and color will be changed.

Button: To trigger the change in font and color, and show a Toast message.

Spinner or Color Picker: To let the user choose a color (optional, or a simple button for a predefined color change).

RadioButtons or a Dropdown Menu to select different font styles (optional).

3. Declare Variables: o Create references for the TextView, Button, Color Picker (if used), and Radio Buttons (if used to change font style).

o Define variables for the font style, color, and the message to display in the Toast.

4. Set Event Listener for the Button: o Set an OnClickListener for the Button. o Inside the onClick() method, write the logic to:

Change the font of the TextView based on the selected font style.

Change the color of the TextView based on the selected color.

Display a Toast message indicating the action has been performed.

5. Implement Font Style Changes: o Change the TextView's font using setTypeface() to apply a selected font style (e.g., bold, italic).

o You can either use predefined styles (like Typeface.BOLD or Typeface.ITALIC) or let the user choose from a list of font styles.

6. Implement Color Change:

o Change the TextView's text color by using the setTextColor() method and applying the selected color (e.g., Color.RED, Color.BLUE, or a color from the Color Picker). o If using a color picker or dropdown, get the selected color and set it on the TextView.

7. Show Toast Message:

o Display a Toast message to the user when the button is clicked, informing them that the text style or color has changed. You can also show a message like (Font and color changed) or any other appropriate message.

8. End the Operation:

o After the changes are made, the application will display the updated TextView with the new font and color, and the Toast message will appear to provide feedback.

PROGRAM:

Step 1: Create the Project

Step 2:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#ffffff"
    android:padding="20dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Font and Color Changer"
        android:textSize="20sp"
        android:textStyle="bold"
        android:gravity="center"
        android:paddingBottom="20dp"
```

```
    android:textColor="#222222"/>

<!-- Display Text -->
<TextView
    android:id="@+id/tvDisplay"
    android:layout_width="match_parent"
    android:layout_height="120dp"
    android:text="Hello, Android!"
    android:textSize="28sp"
    android:gravity="center"
    android:background="#f0f0f0"
    android:textColor="#000000"
    android:layout_marginBottom="24dp"/>

<!-- Color Section -->
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Change Color"
    android:textSize="15sp"
    android:textStyle="bold"
    android:textColor="#333333"
    android:paddingBottom="8dp"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="20dp">

    <Button
```

```
        android:id="@+id/btnRed"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Red"
        android:backgroundTint="#e53935"
        android:textColor="#ffffff"
        android:layout_marginEnd="4dp"/>

    <Button
        android:id="@+id/btnGreen"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Green"
        android:backgroundTint="#43a047"
        android:textColor="#ffffff"
        android:layout_marginEnd="4dp"/>

    <Button
        android:id="@+id/btnBlue"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Blue"
        android:backgroundTint="#1e88e5"
        android:textColor="#ffffff"/>

</LinearLayout>

<!-- Font Section -->
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Change Font Style"
    android:textSize="15sp"
    android:textStyle="bold"
    android:textColor="#333333"
    android:paddingBottom="8dp"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="24dp">

    <Button
        android:id="@+id/btnNormal"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Normal"
        android:layout_marginEnd="4dp"/>

    <Button
        android:id="@+id/btnBold"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Bold"
        android:layout_marginEnd="4dp"/>
```

```
    <Button
        android:id="@+id/btnItalic"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Italic"/>

  </LinearLayout>

  <!-- Toast Button -->
  <Button
    android:id="@+id/btnToast"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Show Toast Message"
    android:backgroundTint="#ff9f0a"
    android:textColor="#ffffff"
    android:textSize="16sp"/>

</LinearLayout>
```

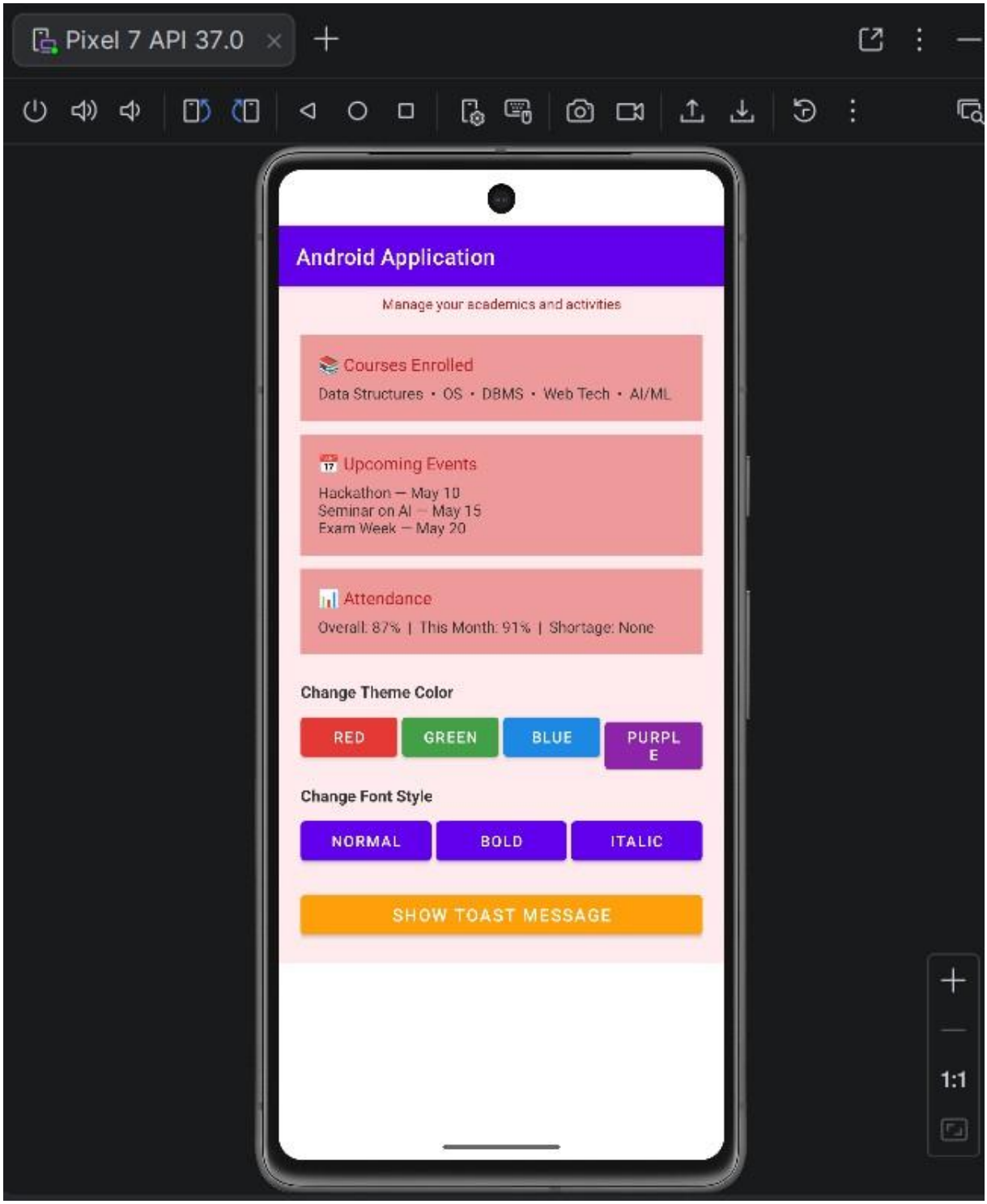
Step 3: Main_Activity.java:

```
package com.example.hostel_management_system;

import android.graphics.Typeface;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

  TextView tvDisplay;
```

```
    Button btnRed, btnGreen, btnBlue;
    Button btnNormal, btnBold, btnItalic;
    Button btnToast;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        tvDisplay = findViewById(R.id.tvDisplay);
        btnRed    = findViewById(R.id.btnRed);
        btnGreen  = findViewById(R.id.btnGreen);
        btnBlue   = findViewById(R.id.btnBlue);
        btnNormal = findViewById(R.id.btnNormal);
        btnBold   = findViewById(R.id.btnBold);
        btnItalic = findViewById(R.id.btnItalic);
        btnToast  = findViewById(R.id.btnToast);

        // Color buttons
        btnRed.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                tvDisplay.setTextColor(0xFFE53935);
                Toast.makeText(SecondActivity.this, "Color changed to Red", Toast.LENGTH_SHORT).show();
            }
        });

        btnGreen.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                tvDisplay.setTextColor(0xFF43A047);
                Toast.makeText(SecondActivity.this, "Color changed to Green",
Toast.LENGTH_SHORT).show();
            }
        });

        btnBlue.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                tvDisplay.setTextColor(0xFF1E88E5);
                Toast.makeText(SecondActivity.this, "Color changed to Blue", Toast.LENGTH_SHORT).show();
            }
```

```
        });

        // Font buttons
        btnNormal.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                tvDisplay.setTypeface(Typeface.DEFAULT);
                Toast.makeText(SecondActivity.this, "Font set to Normal", Toast.LENGTH_SHORT).show();
            }
        });

        btnBold.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                tvDisplay.setTypeface(Typeface.DEFAULT_BOLD);
                Toast.makeText(SecondActivity.this, "Font set to Bold", Toast.LENGTH_SHORT).show();
            }
        });

        btnItalic.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                tvDisplay.setTypeface(Typeface.defaultFromStyle(Typeface.ITALIC));
                Toast.makeText(SecondActivity.this, "Font set to Italic", Toast.LENGTH_SHORT).show();
            }
        });

        // Toast button
        btnToast.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Toast.makeText(SecondActivity.this,
                    "Font & Color Changer App is running!",
                    Toast.LENGTH_LONG).show();
            }
        });
    }
}
```

OUTPUT:



Result: An mobile application is developed in Android studio to change the font and color of the text and display toast message when the user presses the button.